

A method of calculating galaxy infall velocity that can be applied to SDSS data

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The peculiar velocity dispersion profile of galaxies in a dark matter halo depends on the halo density profile, concentration-mass relation and halo mass. Therefore, observations of this profile could be used to verify the validity of current models of halo density profile and concentration-mass relation. However, velocity dispersion profiles have never been directly observed from peculiar velocity observations. My research attempted to observe this profile in simulations, using methods that can be applied to data from the Sloan Digital Sky Survey peculiar velocity catalogue.

Introduction In the currently accepted model of the universe, the Λ CDM model, the universe is made up of dark energy, (cold) dark matter and baryonic matter. Both dark energy and dark matter interact with baryonic matter through gravity only. Shortly after the Big Bang, dark matter collapsed under gravity to form halos. Due to the gravitational interaction between dark matter and baryonic matter, galaxies mostly form in and around dark matter halos [1–3].

There are many different ways to define the boundary of a dark matter halo, and there is much debate over which boundary definition is the most useful [4–8]. One of the most commonly used boundary radii is the virial radius, defined as the radius within which the virial theorem, $\langle T \rangle = -(1/2)\langle U \rangle$ holds true. In practice, this is the radius within which the average mass density is 200 times the critical density of the universe. It only depends on mass, and can be calculated using Eq. (1) [3].

$$r_{\text{vir}} = \left(\frac{GM}{100H_0^2} \right)^{\frac{1}{3}} \quad (1)$$

Whilst the virial radius is usually a good definition for the boundary of a halo, there is an argument that the velocity profile of a halo extends past the virial radius. This motivates the definition of other possible boundary radii based on features of the infall velocity profile [4–8]. Infall velocity is defined as the radial component of a galaxy’s peculiar velocity. Though a galaxy has a velocity associated with the expansion of the universe, it also has a velocity associated with gravitational interactions with other celestial objects, called peculiar velocity [3].

The infall velocity dispersion profile of galaxies depends on the halo density profile and concentration-mass relation, along with the mass of the parent halo. Fig. 1 shows the profile for various halo masses using the Navarro-Frenk-White (NFW) density profile [9] and the concentration-mass relation described by Prada [10]. The amplitude of the velocity dispersion profile will vary for different halo masses, halo density profiles, and concentration-mass relations. Therefore, observations of the velocity dispersion profile of galaxies can be used to verify the validity of the NFW profile and the concentration-mass relation from Prada [10].

My research aimed to observe the infall velocity dis-

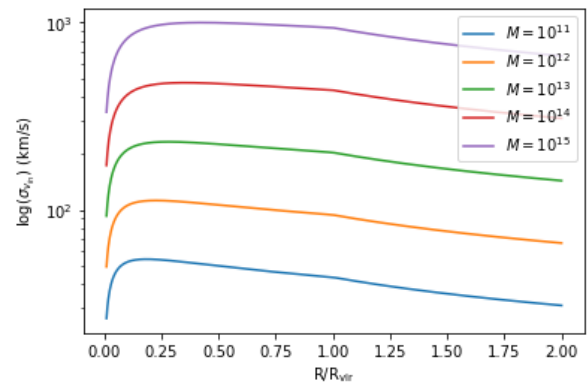


FIG. 1: Plot of the theoretical infall velocity dispersion using an NFW profile [9] and concentration-mass relation described by Prada [10]. On the x-axis is the distance from the halo centre divided by the virial radius. The y-axis uses a logarithmic scale. Each curve shows the profile for a different halo mass.

persion profile of galaxies in simulations using methods that can be applied to real data from the Sloan Digital Sky Survey (SDSS) peculiar velocity catalogue.

Methods The SDSS peculiar velocity catalogue contains measurements of positions and peculiar velocities for 34,059 galaxies. The position of each galaxy is recorded in the form of a redshift, right ascension (RA) and declination (Dec). The redshift can be used to find the distance from the observer (Earth) to the galaxy, using the astropy FlatLambdaCDM comoving_distance function in python. The RA and Dec represent the angular position of the galaxy on the sky, as depicted in Fig. 2. The catalogue also includes the logarithmic distance ratio, $\eta = \log(d(z_{\text{group}})/d(\bar{z}))$. Here, z_{group} is the redshift of the halo centre, and $\bar{z}$ is the cosmological redshift of the galaxy. The peculiar velocity can then be calculated using Eqs. (2) and (3) [11]. It is important to note that this is the component of the peculiar velocity along the line of sight, as shown in Fig. 3.

$$v_p = \frac{cz_{\text{mod}}}{1 + z_{\text{mod}}} \ln(10)\eta \quad (2)$$

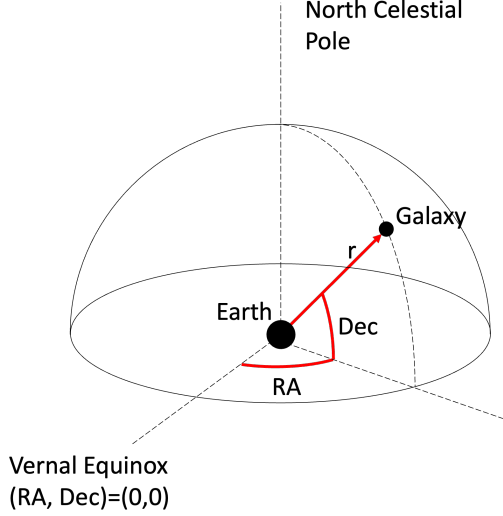


FIG. 2: Diagram of celestial coordinates. r is the radial distance from the Earth to the galaxy, RA is the right ascension, or angle between the vernal equinox and the galaxy, and Dec is the declination, or angle between the north celestial pole and the galaxy.

$$z_{\text{mod}} = z_{\text{CMB}} \left[1 + \frac{1}{2} (1 - q_0) z_{\text{CMB}} - \frac{1}{6} (j_0 - q_0 - 3q_0^2 + 1) z_{\text{CMB}}^2 \right] \quad (3)$$

To find the infall velocity for each galaxy, we also need the position and peculiar velocity of each halo centre. These can be obtained from a catalogue published by Tempel in 2017. This catalogue contains the same parameters as the SDSS catalogue. Both catalogues also contain a group ID, which can be used to match each galaxy to its parent halo [12].

Alongside the SDSS peculiar velocity data, an ensemble of 2,048 simulations was also published. Each of these simulations contain the position and peculiar velocity of each galaxy and halo centre in cartesian coordinates. The simulations also contain RA and Dec, along with a measured redshift and log-distance ratio, and a true redshift and log-distance ratio. It is important to note that the velocity in cartesian coordinates is the exact peculiar velocity, whereas that calculated from the redshift and log-distance ratio is the peculiar velocity along the line of sight [11].

Since the simulations contain three different sets of coordinates for the galaxy peculiar velocity, there are multiple possible methods for finding the infall velocity.

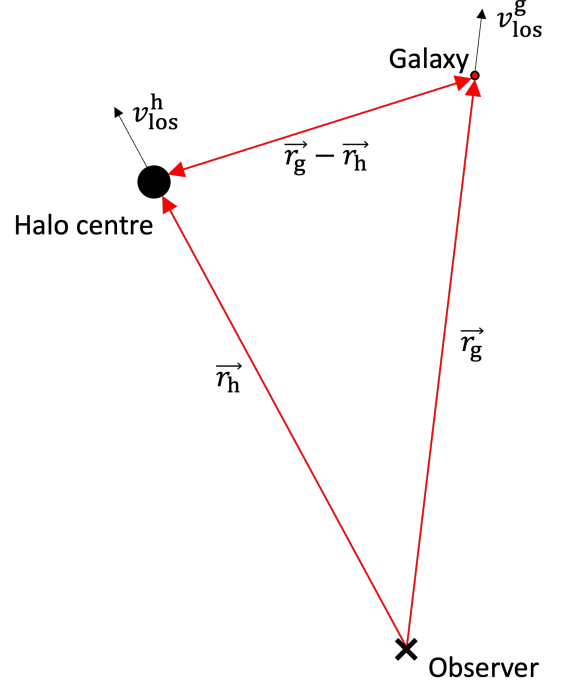


FIG. 3: Diagram of the motion of a galaxy and its halo centre with respect to an observer. v_{los}^g and v_{los}^h are the line of sight velocities of the galaxy and halo centre, respectively. r_g and r_h are the distances from the observer to the galaxy and halo centre respectively. $\vec{r}_g - \vec{r}_h$ is the distance between the galaxy and halo centre.

I used four different methods.

The most obvious way to calculate the infall velocity is using simple vector projection, as outlined in Eq. (4).

$$v_{\text{in},1} = (\vec{v}_g - \vec{v}_h) \cdot \frac{(\vec{r}_g - \vec{r}_h)}{|\vec{r}_g - \vec{r}_h|} = \frac{(v_x^g - v_x^h)(x_g - x_h) + (v_y^g - v_y^h)(y_g - y_h) + (v_z^g - v_z^h)(z_g - z_h)}{\sqrt{(x_g - x_h)^2 + (y_g - y_h)^2 + (z_g - z_h)^2}} \quad (4)$$

This equation will give the exact infall velocity, but it cannot be applied to real data, as the real data only includes the line of sight velocity. The second method I used is to find the line of sight velocity of both the galaxy and halo centre from the cartesian velocity, using Eq. (5).

$$v_{\text{los}} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z} \quad (5)$$

I then took the line of sight velocity for the galaxy and the halo centre and found the infall velocity using Eq. (6)

$$v_{\text{infall}} = \frac{v_{\text{los}}^g |r_g| + v_{\text{los}}^h |r_h| - (v_{\text{los}}^g |r_h| + v_{\text{los}}^h |r_g|) [\cos \delta_h \cos \delta_g \cos(\alpha_g - \alpha_h) + \sin \delta_h \sin \delta_g]}{[|r_g|^2 + |r_h|^2 - 2r_g r_h (\cos \delta_g \cos \delta_h \cos(\alpha_g - \alpha_h) + \sin \delta_h \sin \delta_g)]^{\frac{1}{2}}} \quad (6)$$

Here, α is the right ascension (RA), δ is the declination, r is the distance from the observer and v_{los} is the line of sight velocity. The subscript g refers to the galaxy and the subscript h refers to the halo centre.

Since the simulations don't include the RA and declination for the halo centre, I calculated these from the cartesian coordinates using Eqs. (7) and (8). I also calculated the distance from the observer for both the galaxy and halo centre using Eq. (9).

$$\alpha = \arctan\left(\frac{x}{y}\right) \quad (7)$$

$$\delta = \arcsin\left(\frac{z}{r}\right) \quad (8)$$

$$r = \sqrt{x^2 + y^2 + z^2} \quad (9)$$

The third method I used was to calculate the line of sight velocity for both the galaxy using the true redshift, true log-distance ratio and Eq. (2). The simulations did not include the redshift or log-distance ratio for the halo centre, so I used Eq. (5) to find the line of sight velocity for the halo centre. I then used Eq. (6) to find the infall velocity. The fourth and last method I used was similar to the third method, but using the measured redshift and log-distance ratio.

To find the infall velocity dispersion, I first divided the galaxies into three groups based on the mass of their parent halo. The mass ranges of the three groups were 10^{13} to 10^{14} , 10^{14} to 10^{15} and 10^{15} to 10^{16} . The simulation didn't include any halos with mass less than 10^{13} or greater than 10^{16} . Here, the mass is in units of $h^{-1} M_{\text{sol}}$. For each mass group, I split the galaxies up into 100 bins based on the distance between the galaxy and its halo centre. Each bin covered an equal range of distances. I then calculated the dispersion for each bin by taking the standard deviation of the infall velocities of all galaxies in the bin.

Results Fig. 4 shows plots of the velocity dispersion using the first three methods outlined above, for the three mass bins. Fig. 4a shows the plot for halo masses from 10^{13} to $10^{14} h^{-1} M_{\text{sol}}$. The computational curves on this plot each have a similar shape, which doesn't match the shape of the theoretical curve. The theoretical curve turns down for very small radii and large radii, but the computational curves do not have this turn down, and actually turn up slightly for large radii. However, the amplitude of the v_1 curve matches that of the theoretical curve. This is promising, as the amplitude of the curve will differ for different halo models. Thus, the amplitude matching simulations suggests that the NFW profile is an accurate model.

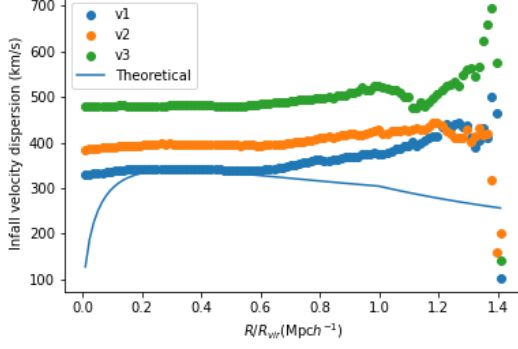
For Fig. 4a, the v_2 and v_3 curves both have amplitudes that are close to that of the v_1 curve. However, for Fig. 4b, which has halo masses from 10^{14} to $10^{15} h^{-1} M_{\text{sol}}$, the three curves match almost exactly. Such a significant difference in the features of the plots for different mass groups is not expected. Therefore, it would be interesting to investigate the reasons why there is such a difference.

Fig. 4c shows the plot for halo masses from 10^{15} to $10^{16} h^{-1} M_{\text{sol}}$. This mass group had much fewer galaxies than the other two, as there were very few galaxies with masses greater than $10^{15} h^{-1} M_{\text{sol}}$. Unlike the other two mass groups, v_1 roughly matches the theoretical curve for most of the radii on the plot. All three computational curves still lack the turn down for very small radii, but they all have similar shapes. Again, the amplitudes of the v_2 and v_3 curves differ slightly from that of the v_1 curve, but they are of the same order of magnitude.

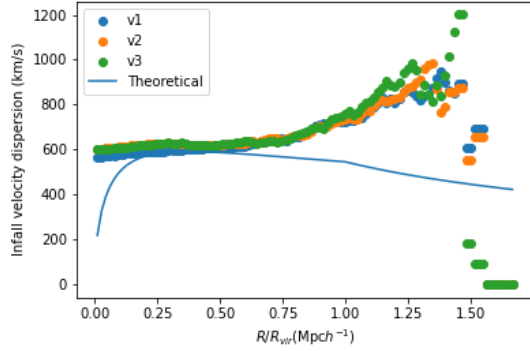
The fact that the amplitudes of the v_2 and v_3 curves are so close to that of the v_1 curve for all three mass bins confirms that the infall velocity equation used (Eq. (6)) is correct. It also means that the line of sight velocity is enough to accurately determine the infall velocity. This suggests that Eq. (6) could be used on real SDSS data.

Fig. 5 shows the infall velocity dispersion for the three mass groups using the measured redshift and log-distance ratio, along with Eqs. (2) and (6). Comparing this plot to Fig. 4, the amplitude of each curve is much greater using the measured redshift and log-distance ratio than for all other methods. This is expected, as the measured peculiar velocity will have a greater uncertainty, resulting in a greater range of velocities in each distance bin. This greater range of velocities results in a higher dispersion.

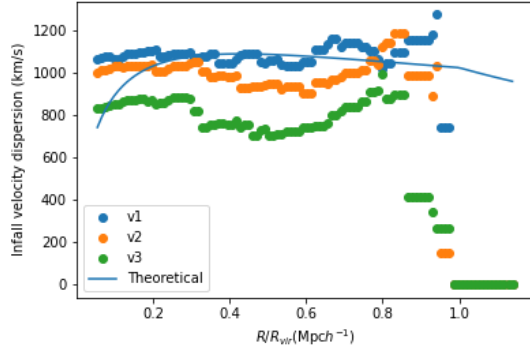
The large difference between the measured infall velocity and the infall velocity calculated using the other three methods suggests that the infall velocity calculated from real data will predict a larger dispersion than the true dispersion. This means that to make meaningful measurements of the infall velocity dispersion, we must develop some way of calibrating the measured dispersion so that it will match the true dispersion.



(a) Halo masses ranging from 10^{13} to $10^{14} h^{-1} M_{\text{sol}}$



(b) Halo masses ranging from 10^{14} to $10^{15} h^{-1} M_{\text{sol}}$



(c) Halo masses ranging from 10^{15} to $10^{16} h^{-1} M_{\text{sol}}$

FIG. 4: Plots of infall velocity dispersion for three different mass ranges, each using three different methods to calculate the infall velocity. The x-axis shows the distance between the halo centre and galaxy divided by the halo's virial radius. The theoretical curve was constructed using an NFW profile, a concentration-mass relation described by Prada and the average mass of all the halos in the mass range. v1 was found using Eq. (4), v2 was found using Eqs. (5) and (6) and v3 was found using Eqs. (2) and (6) with the true redshift and log-distance ratio.

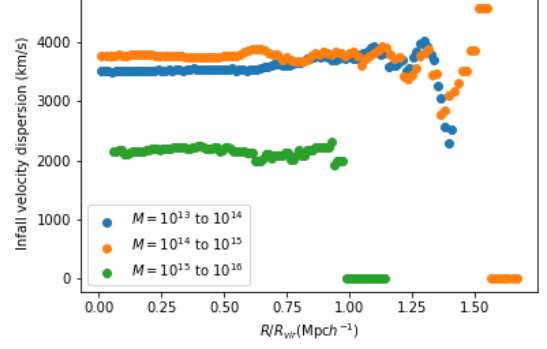


FIG. 5: Plots of infall velocity dispersion for three different mass ranges. The x-axis shows the distance between the halo centre and galaxy divided by the halo's virial radius. The infall velocity was calculated using Eqs. (2) and (6) with the measured redshift and log-distance ratio.

Conclusions and Future Work In conclusion, the amplitude of the infall velocity dispersion profile calculated using the exact position and peculiar velocity agrees with the theoretical curve using an NFW profile [9] and concentration-mass relation from Prada [10]. The two methods using true line of sight velocity are also very close to the theoretical curve, but the deviation from theory depends largely on the halo mass group. Therefore, a possible avenue for future research is to investigate why there is such a significant difference between the different mass groups. This could be done by investigating how the dispersion profiles change if smaller mass groups are used.

The infall velocity dispersion profiles calculated using the measured peculiar velocity have a much higher amplitude than the theoretical curve, as expected. A possible avenue for future work is to develop a calibration method to relate the dispersion profile using the true peculiar velocity to that using the measured peculiar velocity. Then, this calibration method could be applied to velocity dispersion profiles from SDSS data, in order to compare them to the theory.

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